WHAT IF?! EP 10

FREE SURVIVAL PLAN
you commented ANTS

Ant War Survival Plan



Ten ant numbers that will ruin your picnic, a war scoreboard, and a kitchen defence plan. Know the enemy. They have surgeons.

10 ANT NUMBERS

1. **20 quadrillion** ants on Earth — A 2022 PNAS estimate from 489 studies, probably an underestimate.
2. **2.5 million** ants per human — 20 quadrillion divided by 8 billion people. My maths.
3. **12 Mt** dry carbon — Outweighs all wild birds and wild mammals combined. About 20% of human biomass, not equal to it.
4. **6,000 km** one supercolony — An Argentine ant colony from Italy to Spain's Atlantic coast, with no aggression inside it.
5. **200,000** in one raid — Army ants raid in swarms of up to about 200,000 workers (estimates vary).
6. **40 min** one amputation — Carpenter ants amputate an injured nestmate's leg. About 90 to 95% of femur-wound cases survive.
7. **50 million** years farming — Ants farmed fungus about 50 million years ago. Humans started farming about 12,000 years ago.
8. **~250** naps a day — Fire ant workers nap about 250 times a day for about 1.1 minutes: roughly 4.8 hours of sleep.
9. **306 million** workers in one colony — The Ishikari supercolony in Hokkaido: also about 1.08 million queens (old survey, so "estimated").
10. **4.0+** sting pain — The bullet ant sting sits at the very top of Schmidt's sting pain index.

Sources Schultheiss et al., PNAS 2022 (PMC9586285) · Giraud, Pedersen, Keller, PNAS 2002 (PMC122904) · Frank et al., Current Biology 2024 (PubMed 38959879) · Schultz and Brady, PNAS 2008 (PMC2291106) · Cassill et al. 2009, J Insect Behav · Britannica (bullet ant)

The war scoreboard

Four rounds, as in the reel. Who wins?



ROUND 1 NUMBERS



20 quadrillion vs 8 billion. They outweigh every wild bird and wild mammal combined.

ANTS

ROUND 2 SURGEONS



Field hospitals. Amputation takes about 40 minutes and most patients survive. Untreated infected femur wounds: under 40% survived.

ANTS

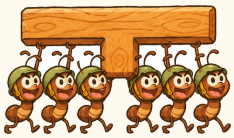
ROUND 3 FARMING



50 million years vs 12,000. Leafcutters do not eat leaves: they farm the fungus. Amateurs, we are.

ANTS

ROUND 4 TEAMWORK



The piano puzzle. Weizmann scientists gave ants and humans the same T-shaped load to move through a maze. Ant groups got better. Human groups did not. Individually, humans beat ants.

ANTS

YOUR KITCHEN DEFENCE PLAN

These are general household common sense, not findings from our research. The ant facts everywhere else are sourced.

1. **Starve the army.** Wipe up crumbs and sticky spills straight away.
2. **Seal the supplies.** Keep sweet and sticky food in closed containers.
3. **Check the warm gear.** Rasberry crazy ants (first identified in Houston, 2002) pile into warm electrical equipment and cause short circuits and stuck switches.

Living rafts

Fire ants link their bodies into floating rafts to survive floods (PNAS 2011). An army that builds its own boat.

Venom, handle with care

Maricopa harvester ant venom is the most toxic insect venom measured, in mice (LD50 0.12 mg/kg). Do not extrapolate that to people.

Honest verdict

We would lose, but ants are busy fighting each other: the two big Argentine supercolonies in Europe are at war.



Final score: ants 4, humans 0, crumbs 100. (One human still beats one ant at the puzzle.) Skipped on purpose: "30 million ants die at the border" (no paper found) and "ants lift 50x their weight, so scale them up" (square-cube law says no).

Sources Dreyer et al., PNAS 2024 (Weizmann, EurekAlert) · Frank et al. 2024 · Mlot, Tovey, Hu, PNAS 2011 (rafts) · Wikipedia, LiveScience (Rasberry crazy ant) · Guinness; UF Book of Insect Records (harvester venom) · Higashi and Yamauchi (Ishikari)